# Tech News Tracker

SMAI Assignment 3 · Task T9.3 Tier 1 · IIIT Hyderabad 2025–26

Vidvathama R (2024122002) Snehil Sanjog (2023102051)
Sarvesh Takbhate (2023102039)
Team: **Yoraha Creations** · IIIT Hyderabad

**Abstract.** The rapid growth of technology journalism across TechCrunch, The Verge, and YourStory makes it difficult for users to stay updated efficiently. This project presents an automated tech-news aggregation and summarisation system that collects articles from multiple RSS feeds, classifies them into technology subcategories using a zero-shot NLI model, and generates concise summaries using a large language model. Experimental evaluation on the India Headlines News Dataset shows that the proposed pipeline achieves **90.7% accuracy** on binary tech vs. non-tech classification with **zero fine-tuning**.

**Live Demo:** https://huggingface.co/spaces/SylvesterSsj/tech-news-terminal
**GitHub:** https://github.com/Sylvester4582/Yoraha_Creations_SMAI_A3

## Introduction

Technology news is produced at an extremely high rate across online media platforms. Readers often spend significant time filtering relevant articles from large streams of information. This project builds an intelligent news-tracking application that performs three major tasks:

1. Aggregates live articles from multiple RSS feeds.
2. Classifies articles into meaningful technology subcategories without any labelled training data.
3. Generates concise summaries using a large language model for quick reading.

*Core research question:* How effectively can a zero-shot NLI model generalise to technology-news classification without any fine-tuning? We evaluate on a benchmark derived from the India Headlines News Dataset.

# Dataset

## RSS Feed Sources

Table 1: Live RSS Feed Sources

| Source | Feed URL | Articles/day |
|---|---|---|
| TechCrunch | https://techcrunch.com/feed/ | ∼20 |
| The Verge | https://www.theverge.com/rss/index.xml | ∼30 |
| YourStory | https://yourstory.com/feed | ∼15 |

The system retrieves the 10 most recent articles from each source (30 total) and sorts them by publication timestamp.

## Evaluation Dataset

The **India Headlines News Dataset** (Therohk, Kaggle) contains 21 years of labelled Indian news headlines. We construct a balanced binary benchmark:

- **Positive class:** 150 technology headlines (`headline_category == "tech"`)
- **Negative class:** 150 headlines from entertainment, sports, lifestyle, and world
- **Total:** 300 balanced evaluation examples

# Methodology

## Pipeline Overview

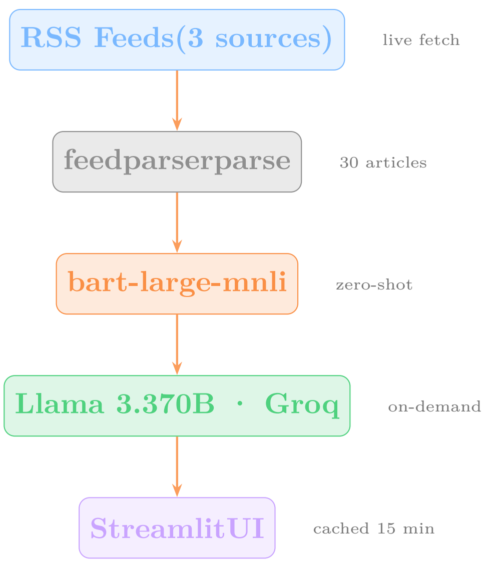


Figure 1: End-to-end application pipeline

### Zero-Shot Classification

**How zero-shot classification works:** Traditional text classifiers require labelled training data for every category. Zero-shot classification removes this requirement by repurposing a Natural Language Inference (NLI) model. NLI models determine whether a *hypothesis* sentence is *entailed by*, *contradicted by*, or *neutral to* a *premise* sentence.

We reformulate classification as an NLI task. For example:

**Premise:** "Apple launches new M4 chip for MacBook Pro lineup."

**Hypothesis:** "This text is about consumer electronics, gadgets, chips, or hardware devices."

**Output:** $P(\text{entailment}) = 0.94 \rightarrow$ Gadgets & Hardware

The label whose hypothesis receives the highest entailment score becomes the predicted category — with no task-specific training required. The NLI inference flow is:

**Article text (title + 300 chars)**

↓

**BART-MNLI (406M)**

↓

$\arg\max_i P(\textbf{ent} \mid \ell_i)$

The five candidate subcategories are:

Table 2: Technology Subcategories and Hypothesis Phrases

| Display Label | Hypothesis phrase fed to model |
|---|---|
| AI & Machine Learning | "artificial intelligence or machine learning" |
| Startups & Business | "technology startups, funding, or business strategy" |
| Gadgets & Hardware | "consumer electronics, gadgets, chips, or hardware devices" |
| Software & Apps | "software applications, platforms, or developer tools" |
| General Tech | "general technology news" |

### Summarisation

Article summaries are generated using **Llama 3.3 70B** via the Groq API. Each summary contains exactly three concise bullet points. Summaries are generated *on-demand* (when the user expands an article) and cached for one hour to minimise API calls.

### Caching Strategy

Table 3: Caching Mechanisms

| Component | Mechanism | TTL |
|---|---|---|
| Model weights | `@st.cache_resource` | Process lifetime |
| RSS fetch + classify | `@st.cache_data` | 15 minutes |
| Generated summaries | `@st.cache_data` | 1 hour |

## Results

### Binary Classification Performance

Table 4: Binary Classification Results (300-example benchmark, CPU)

| Metric | Value |
|---|---|
| Accuracy | **0.907** |
| Precision | 0.912 |
| Recall | 0.900 |
| F1 Score | **0.906** |
| Mean confidence (correct) | 0.881 |
| Mean confidence (wrong) | 0.737 |

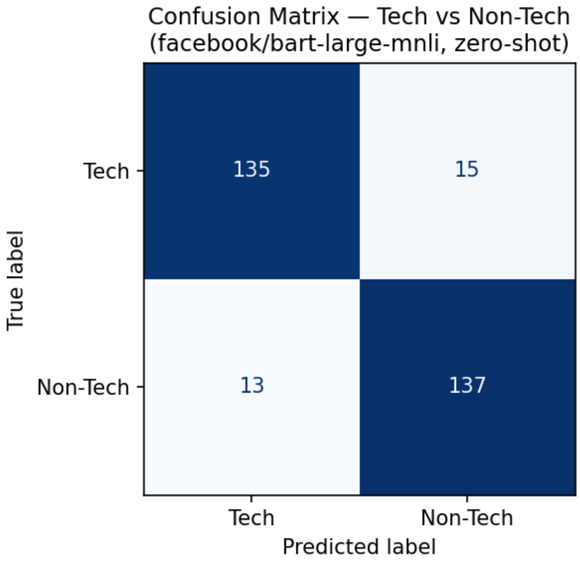


Figure 2: Confusion Matrix — Tech vs. Non-Tech (facebook/bart-large-mnli, zero-shot)

### Confidence Distribution

Correct predictions cluster near high confidence ($> 0.85$), while incorrect predictions are spread more uniformly. This means the model's confidence score is a reliable uncertainty signal — the UI colour-codes each article card accordingly.

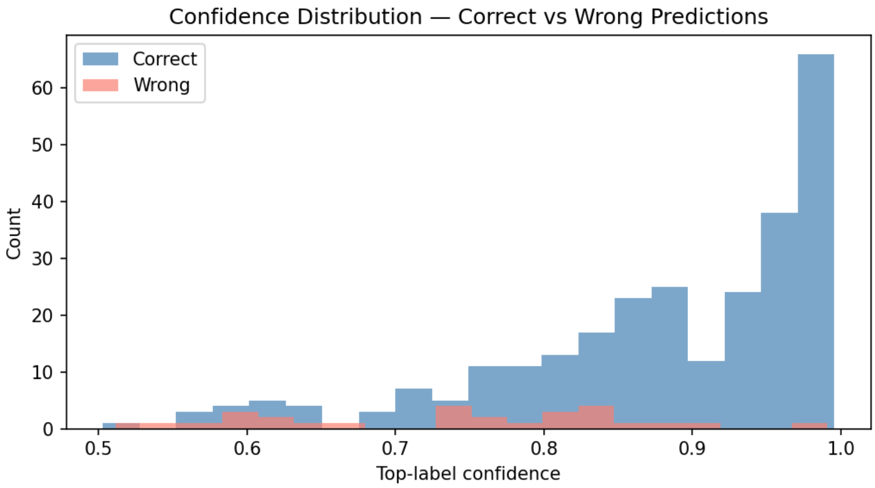


Figure 3: Confidence Distribution — Correct vs. Wrong Predictions

### Subcategory Distribution

Among tech-labelled headlines, Gadgets & Hardware dominates (67/150), consistent with the India Headlines dataset's skew toward device and product coverage.

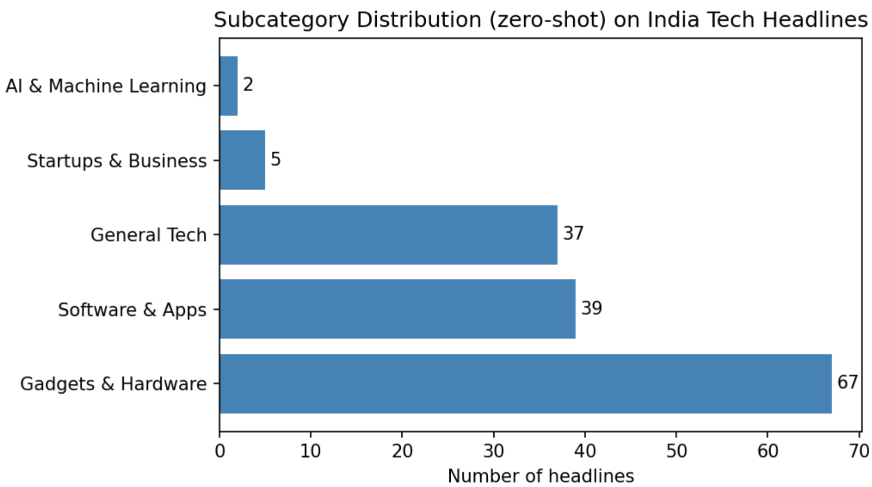


Figure 4: Zero-shot Subcategory Distribution on India Tech Headlines ($n = 150$)

### Pipeline Latency

Latency was measured end-to-end using the actual application modules (`rss_fetcher`, `classifier`, `summarizer`) on CPU, matching exactly what `app.py` does on each page load.

Table 5: Measured Pipeline Latency (CPU, 30 articles)

| Stage | Description | Time |
|---|---|---|
| RSS Fetch | 3 feeds, 30 articles | 2.158 s |
| classify_articles | Batch NLI, 30 articles | 58.936 s |
| per article | (single article, batched) | 1964.5 ms |
| summarize | 1 article via Groq API | 7.833 s |
| **Page load total** | (fetch + classify; summarize is lazy) | **61.094 s** |

**Note:** Classification is the bottleneck on CPU (~59s for 30 articles). On the live HuggingFace Space, the model is cached after the first load and the 15-minute `st.cache_data` TTL means most users never wait for re-classification. GPU inference would reduce classification to ∼3–5s.

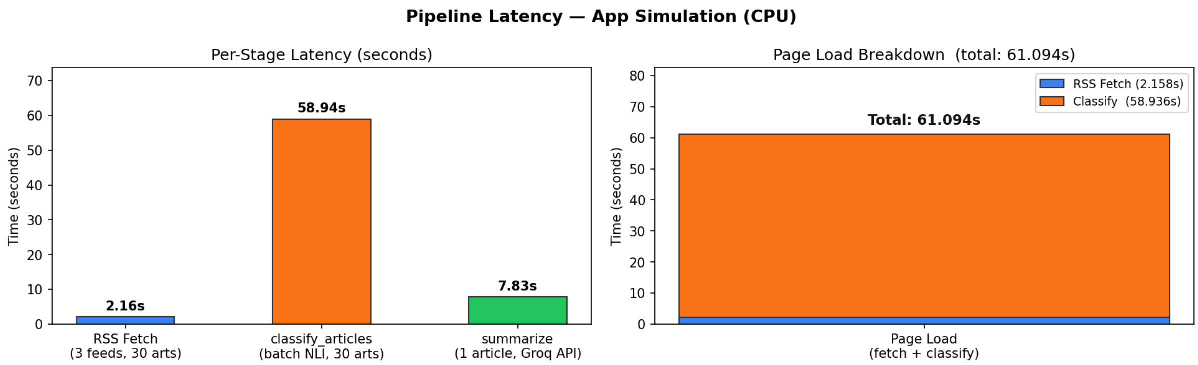


Figure 5: Per-stage latency (left) and page-load breakdown (right) — CPU, app simulation

## Ablation Study

An ablation study was conducted to analyse the effect of label phrasing on zero-shot classification performance. The NLI head is trained on natural-language premise-hypothesis pairs, so well-formed, descriptive label phrases align better with the training distribution than bare keywords.

Table 6: Effect of Label Phrasing on Binary Classification

| Label Style | Positive label example | Accuracy | F1 |
|---|---|---|---|
| **Descriptive** | "technology news" | **0.907** | **0.906** |
| Short keyword | "tech" | 0.647 | 0.727 |
| Hypothesis sentence | "This headline is about technology." | 0.637 | 0.725 |

**Key finding:** Descriptive labels outperform short keywords by **+26.0 pp accuracy** and hypothesis sentences by **+27.0 pp accuracy**. Descriptive natural-language labels are the optimal phrasing strategy for NLI-based zero-shot classifiers.

## Limitations

1. **Domain shift:** The model was trained on general English NLI corpora. Indian startup jargon and Hinglish in YourStory headlines reduce classification confidence.
2. **Summarisation hallucination:** Llama 3.3 70B may generate plausible-sounding but inaccurate bullet points when article content is short or behind a paywall.
3. **CPU latency:** Classification takes ∼59s on CPU for 30 articles. This is mitigated by caching but makes cold starts slow.
4. **RSS availability:** YourStory's feed occasionally times out; the app degrades gracefully using the other two sources.
5. **No fine-tuning:** Zero-shot accuracy of 90.7% is strong but below what a fine-tuned model would achieve (>95%).

## Main Page of UI

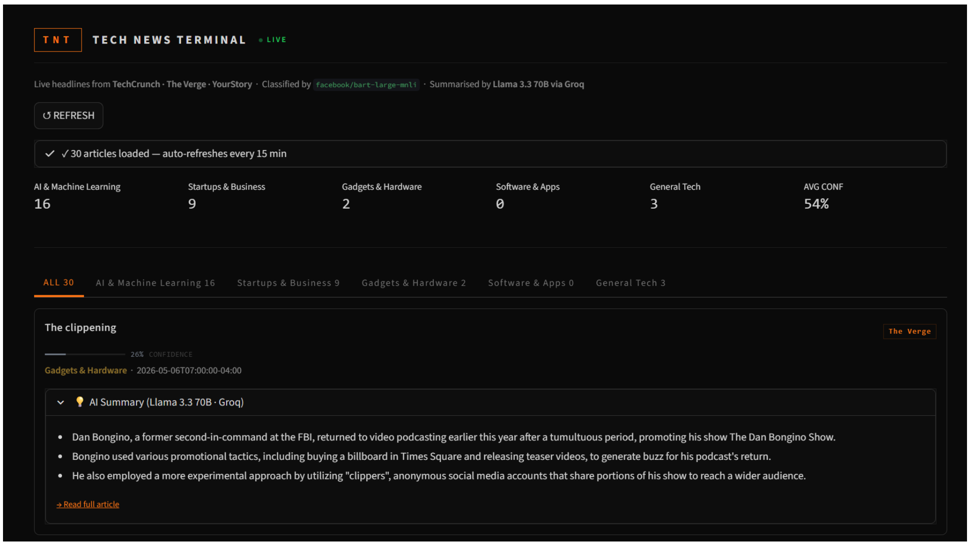


Figure 6: Screenshot of UI

## Conclusion

This project demonstrates that combining RSS aggregation, zero-shot NLI classification, and LLM summarisation produces an effective real-time technology-news tracker. Despite requiring **no task-specific fine-tuning**, the system achieves **90.7% accuracy** on a balanced benchmark, and ablation experiments confirm that descriptive label phrasing is critical for NLI-based zero-shot performance.

## References

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